STPM Physics · Semester 2 · Chapter 15

ELECTRIC CURRENT

15.1 Conduction of Electricity

DEFINITIONS & LAWS

- **Electric current** is the rate of flow of charge through a cross-section of a conductor.
- **Conduction in a metal** is due to the free (delocalised) electrons, which have large random thermal velocities in all directions; an applied p.d. sets up an electric field that superimposes a small net drift velocity opposite to the field.
- **Conventional current** is taken in the direction of flow of positive charge, which is opposite to the direction of drift of the electrons.
- **Explain why the resistance of a metal increases with temperature:** the lattice ions vibrate with greater amplitude, so the free electrons collide with them more frequently; the mean free time between collisions decreases while the free electron density is almost constant.
- **Explain why the conductivity of a semiconductor increases with temperature:** more electrons are released from the covalent bonds, producing more electron-hole pairs; the increase in the number of charge carriers outweighs the increased lattice scattering.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Electric current	$I = \frac{Q}{t}$	I = current (A, where A = C s^{-1}); Q = charge passing a point (C); t = time (s)
Charge and number of electrons	$Q = Ne$	N = number of electrons; e = 1.60×10^{-19} C
Current and drift velocity	$I = nAev_d$	n = free electron density (m^{-3}); A = cross-sectional area (m^2); v_d = drift velocity (m s^{-1})

- Typical values: drift velocity ~10^{-4} m s^{-1}; random thermal speed ~10^5 to 10^6 m s^{-1}.

TIPS TO ANSWER

- For 'explain' questions, state the three linked points: free electrons, collisions with lattice ions, and the effect on the mean free time / drift velocity.
- State that the electrons drift in the direction opposite to the electric field and to the conventional current.

- For charge-counting problems: Q = It, then N = $\frac{Q}{e}$, with time converted to seconds and the answer in standard form.
- When contrasting metals and semiconductors, compare the change in carrier density with the change in scattering.

SECTION A TRAPS

Trap: The resistance of a metal increases because the number of free electrons falls with temperature. The free electron density is nearly constant; it is the collision rate that increases.

Trap: Forgetting to convert minutes to seconds in Q = It, giving an answer 60 times too small.

Trap: Stating that the free electrons move in the direction of the conventional current; they drift in the opposite direction.

TRICKS

Trick: Metal: n constant, collisions increase, so R increases. Semiconductor: n increases faster than the scattering, so R decreases. Eliminate any option that changes the charge e of the carrier.

15.2 Drift Velocity

DEFINITIONS & LAWS

- **Drift velocity** is the **mean** velocity of the free electrons along the conductor due to the applied electric field, superimposed on their random thermal motion.
- **Why a lamp lights almost instantaneously:** the electric field is established along the whole circuit at a speed close to that of light, so the free electrons throughout the wire, including those in the filament, begin to drift at once.
- **In a steady current** the drift velocity is the same at every point of a uniform wire; in a wire of varying cross-section it is greater where the area is smaller.
- **In a semiconductor** both electrons and holes contribute to the current; they drift in opposite directions but their contributions to the conventional current add.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Current and drift velocity	$I = nAev_d$	n = number of free charge carriers per unit volume (m^{-3}); A = cross-sectional area (m^2); e = 1.60×10^{-19} C; v_d = drift velocity (m s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Drift velocity	$v_d = \frac{I}{nAe}$	I = current (A)
Proportionality of drift velocity	$v_d \propto I$; $v_d \propto \frac{1}{nA}$	Each holds with the other quantities constant
Circular cross-sectional area	$A = \pi r^2 = \frac{\pi d^2}{4}$	r = radius (m); d = diameter (m); for a circular cross-section
Current in a semiconductor	$I = neA(v_e + v_h)$	v_e = electron drift velocity (m s^{-1}); v_h = hole drift velocity (m s^{-1})

TIPS TO ANSWER

- State the formula I = nAevd, then rearrange for the unknown before substituting, with every quantity in SI units.
- Find A from the diameter first: $A = \frac{\pi d^2}{4}$, with d converted to metres (1 mm = 10^{-3} m).
- For ratio questions write $v_d \propto \frac{1}{A}$ (same material and current) and substitute the area ratio.
- Give the answer in m s^{-1} in standard form to 3 s.f.; the typical value is of order 10^{-4} m s^{-1}, which serves as a check.
- Define drift velocity as the **mean** velocity of the free electrons; leaving out "mean" loses the mark.
- In $\sigma = \frac{ne^2t}{m}$, t is the mean free time (mean time between collisions), as in A-level reference books, not the time for an electron to travel from one end of the wire to the other.

SECTION A TRAPS

Trap: Using the radius as the diameter, or the reverse, in $A = \frac{\pi d^2}{4}$; this gives an error of a factor of 4.

Trap: Halving the diameter gives double the drift velocity. The area falls to one quarter, so vd becomes 4 times larger.

Trap: Believing that the electrons themselves travel at nearly the speed of light; only the electric field is established at that speed.

Trap: Writing $v_d \propto n$; for fixed I and A, vd is inversely proportional to n.

TRICKS

> **Trick:** Same I, same material: vd $\propto \frac{1}{A} \propto \frac{1}{d^2}$. Halve d, vd × 4; double d, vd ÷ 4.

> **Trick:** A numerical check: copper wires carrying a few amperes give vd between 10^{-5} and 10^{-3} m s^{-1}; an answer far from this range signals a wrong power of ten.

15.3 Current Density

DEFINITIONS & LAWS

- **Current density** is the current per unit cross-sectional area perpendicular to the direction of the current at that point, $J = \frac{I}{A}$.
- **Current density** is a vector quantity with the direction of the conventional current.
- **In a steady current** the current I is the same at every cross-section of a conductor, but the current density is greater where the cross-sectional area is smaller.
- **Ohm's law (microscopic form):** the current density in a conductor is directly proportional to the electric field strength, $J = \sigma E$, at constant temperature.
- **Analogy:** electric current corresponds to the rate of heat flow, and potential difference to temperature difference; conductivity corresponds to thermal conductivity.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Current density	$J = \frac{I}{A}$	J = current density (A m^{-2}); I = current (A); A = cross-sectional area (m^2)
Current density and drift velocity	$J = nev_d$	n = free electron density (m^{-3}); e = 1.60×10^{-19} C; v_d = drift velocity (m s^{-1})
Current density and field	$J = \sigma E$	σ = electrical conductivity (Ω^{-1} m^{-1} or S m^{-1}); E = electric field strength (V m^{-1})
Field in a uniform wire	$E = \frac{V}{L}$	V = p.d. across the wire (V); L = length of wire (m); for a uniform wire

TIPS TO ANSWER

- Convert the area to m^2 before substituting: 1 mm^2 = 10^{-6} m^2, 1 cm^2 = 10^{-4} m^2.
- Write $J = \frac{I}{A}$, substitute with units, then state the answer in A m^{-2} in standard form.
- For a circular wire, calculate $A = \pi r^2$ or $\frac{\pi d^2}{4}$ first and show this step.

- For tapering conductors state that I is constant while $J \propto \frac{1}{A}$, so J is largest at the narrowest section.

SECTION A TRAPS

Trap: Current density is the same at all points because the current is the same; the area differs, so $J = \frac{I}{A}$ differs.

Trap: Mixing mm^2 and m^2, for example 10^{-3} instead of 10^{-6} for mm^2 to m^2; the answer is then wrong by a factor of 1000.

Trap: Area reduced to one third gives J three times larger, not nine times; J is inversely proportional to A, not to A^2.

TRICKS

Trick: $J \propto \frac{1}{A}$ at constant current: area ÷ 3 gives J × 3.

Trick: Use units to check: A m^{-2} requires the area in m^2; for a 1 mm^2 wire carrying 1 A, J = 10^6 A m^{-2}, a useful benchmark.

15.4 Electric Conductivity and Resistivity

DEFINITIONS & LAWS

- **Resistivity** of a material is the resistance of a wire of that material of unit length and unit cross-sectional area, $\rho = \frac{RA}{L}$ (unit: Ω m).
- **Conductivity** is the reciprocal of resistivity, $\sigma = \frac{1}{\rho}$ (unit: Ω^{-1} m^{-1} or S m^{-1}); both are properties of the material, not of its size or shape.
- **Conductance** is the reciprocal of resistance, $G = \frac{1}{R}$ (unit: siemens, S = Ω^{-1}).
- **Ohm's law**: the steady current in a metallic conductor is directly proportional to the potential difference across it, provided the physical conditions (such as temperature) remain constant.
- A **superconductor** is a material whose resistivity becomes zero at and below its critical temperature T_c; **superconductivity** is this phenomenon of zero resistivity below the critical temperature. Uses: strong electromagnets (MRI, particle accelerators), maglev trains and lossless power transmission.
- **Resistance** is the ratio of the potential difference across a conductor to the current through it, $R = \frac{V}{I}$.

- **Variation with temperature:** for a metal the resistivity increases with temperature because lattice vibrations increase the collision rate; for a semiconductor the resistivity decreases with temperature because the number of charge carriers increases rapidly.
- **Stretching a wire:** at constant volume, if the length is multiplied by k, the area is divided by k, so the resistance is multiplied by k^2.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Resistance of a conductor	$R = \frac{\rho L}{A}$	R = resistance (Ω); ρ = resistivity (Ω m); L = length (m); A = cross-sectional area (m^2)
Conductivity	$\sigma = \frac{1}{\rho} = \frac{L}{RA}$	σ = conductivity ($\Omega^{-1}\ m^{-1}$ or $S\ m^{-1}$)
Resistance at constant volume	$V = AL;\ R \propto L^2$ (or $R \propto \frac{1}{A^2}$)	V = volume (m^3), constant; L = length (m); A = cross-sectional area (m^2)
Relaxation time	$\tau = \frac{m}{ne^2\rho}$	τ = relaxation time (s); m = electron mass (9.11×10^{-31} kg); n = free electron density (m^{-3}); e = 1.60×10^{-19} C; ρ = resistivity (Ω m)
Power dissipated in a resistor	$P = I^2R = \frac{V^2}{R}$	P = power (W); I = current (A); V = p.d. (V); R = resistance (Ω); combine with $R = \frac{\rho L}{A}$ in heater-coil problems

TIPS TO ANSWER

- State $R = \frac{\rho L}{A}$, convert the diameter to metres and calculate $A = \frac{\pi d^2}{4}$ before substituting.
- For ratio problems write each resistance as $\frac{\rho L}{A}$, cancel the common quantities and substitute the length and area ratios.
- For a reshaped wire state explicitly that volume is constant, so A changes when L changes.
- State the final answer in Ω (or $\Omega^{-1}\ m^{-1}$ for σ) to 3 s.f. with the unit.
- Use $\sigma = \frac{ne^2t}{m}$ to explain: conductor vs semiconductor (much larger n in a conductor); conductor heated: atoms vibrate with larger amplitude, the mean time between collisions t decreases, σ decreases and ρ increases; semiconductor heated: n increases, σ increases and ρ decreases.
- Resistivity depends only on the material (and temperature): cutting a wire to half its length halves R, but ρ is unchanged.

SECTION A TRAPS

Trap: Stretching to twice the length gives 2R. The area also halves at constant volume, so the resistance becomes 4R.

Trap: Using cm^2 as m^2: 0.040 cm^2 is 4.0×10^{-6} m^2, not 4.0×10^{-4} m^2.

Trap: Using the diameter as the radius in $A = \pi r^2$, which gives A four times too large and R four times too small.

Trap: Treating conductivity as dependent on the length and area of the sample; it is a property of the material only.

TRICKS

Trick: Two materials with identical dimensions: $\sigma \propto \frac{1}{\rho}$, so doubling ρ halves σ.

Trick: Same material: $R \propto \frac{L}{A} \propto \frac{L}{d^2}$. Length × 2 with area ÷ 2 gives R × 4; length × 2 with diameter ÷ 2 gives R × 2 × 4 = 8 times. Check whether the question gives the area or the diameter.